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Regeneration of artificial injuries on scleractinian corals and coral/algal competition for newly formed substrate

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Abstract

Porites cylindrica and *Porites lutea* fragments of colonies were inflicted with five different injury types: chisel, file, Water Pik, osmotic and cement injuries. The fragments were maintained in outdoor aquaria for a period of 240 days under light intensities varying from 2–5% to 70–90% of incident surface photosynthetic active radiation (PAR_0). During the exposure, changes in weight of the fragments, the rates of regeneration of the injuries, abundance of algae and animals settled onto injured areas were monitored. The regeneration rate of the injuries depended on interspecific differences in corals, injury types, number and composition of algae and animals settled onto the lesions, and light and temperature conditions. Competitive interactions between polyps and settlers occurred after colonizers settled onto the damaged surface or the live tissue. It is noteworthy that recovered coral tissue generally overgrew about 100 algal species with or without inhibition of coral growth by algae. In the summer period, the cyanobacterium *Lyngbya majuscula* covered some lesions (osmotic and cement) by 100%, thus reducing dramatically the regeneration rate of the inflicted injuries and also caused coral bleaching when in direct contact.
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Keywords: Coral/algal competition; Injury; Overgrowth; *Porites cylindrica*; *Porites lutea*; Regeneration

1. Introduction

Competition between scleractinian corals and benthic algae is considered fundamental to the overall status of coral reefs. Competition is an important process determining the structure, composition and the abundance of both coral and algal benthic communities on coral reefs (Littler and Littler, 1984; Lapointe, 1989; Hughes, 1994; Miller, 1998; McCook et al., 2001; Diaz-Pulido and McCook, 2002). McCook et al. (2001) have suggested six mechanisms of coral/algal competition: overgrowth, shading, abrasion, chemical, pre-emption/recruitment barrier, and epithelial sloughing. Corals and algae are

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interspecific in competition for space, substrate and resources (McCook et al., 2001). In most cases, external factors, especially their sharp changes, determine not only the abundance and structure of both communities but also the superiority over the rest (McCook et al., 2001; Diaz-Pulido and McCook, 2002; Lapointe, 1989; Done, 1992).

The competition between coral polyps and algae for substratum, space and sources is a common and constant process on coral reefs with varying success for the competitors. However, for the past decades owing to global natural catastrophes, degradation of coral reefs and the so-called phase shifts from coral to plant dominance have been observed. Plant reefs consist mostly of algal and seagrasses grown on newly formed substrate due to dead and damaged colonies of hermatypic corals. The competition between corals and algae begins immediately after the formation of this substrate. Micro- and macro-algae, cyanobacteria and seagrasses as well as benthic invertebrate animals including settled planulae and survived coral polyps, participate in this competition. During the first 1–2 years after catastrophe, algae are the winners and occupy more than 90% of the newly formed substrate (Lapointe, 1989; Loya et al., 2001). It is known that during the first years, after the catastrophe, recovery of coral reef on the new substrate also occurs by the formation of new colonies (Sakai et al., 2004), by lesion healing and by spreading of polyps on the previously damaged colonies (Titlyanov, unpublished data).

The ability of many corals to recover from damage, depending on the position, size, shape and type of the damage, has been well documented (e.g., Bak and Steward-van Es, 1980; Bak, 1983; Wahle, 1983; Rinkevich and Loya, 1989; Meesters et al., 1992, 1994; Meesters and Bak, 1993, 1995; Hall, 1997, 2001; Oren et al., 1997; Marshall, 2000). However, to the best of our knowledge, the influence of the settled organisms (algae and animals) on the recovery of injuries, especially in relation to light intensity has largely been ignored (McCook et al., 2001).

In this paper, we show results of some experiments on the study of the dynamics and mechanisms of competition for newly formed substrate (injured surface of colonies) between communities of coral polyps and micro- or macro-algae or some animal organisms settled onto the damaged surface of scleractinian corals. In this study, we addressed the following hypothetical ideas:

1) Competitive success of coral for newly formed substrate (injured part of colony) depends on the physiological state of the coral, environmental conditions, especially light intensity, and also on the density and the structure of algal and animal communities colonizing the substrate.
2) Competitive success of the algae for newly formed substrate depends on the number of spores and thalli fragments settled onto the injury, the composition of algal community, the duration of the existence of the algal community, the physiological state and competitive abilities of coral polyps and environmental conditions.

The major objectives were to examine how scleractinian corals respond to different types of inflicted injuries and to investigate the impact of a given injury on species with different morphologies by monitoring their regenerative abilities to the respective injury under different light conditions. A second major objective was to monitor settlement on the injured areas by algae in various seasons and under different light and temperature conditions.

2. Materials and methods

2.1. Study site, time and conditions

All experiments were conducted in outdoor aquaria (50 L each) at the Sesoko Station of Tropical Biosphere Research Center of the University of the Ryukyus (Sesoko Island, Okinawa, Japan, 26°38′N, 127°52′ E) from November 2002 to August 2003. In the winter and spring periods, water temperature in the aquaria was 20–26 °C and 18–24 °C during the daytime and at night, respectively. In the summer season, the temperature was 28–31 °C and 26–28 °C during the daytime and at night, respectively. Mineral nutrition and zooplankton concentration in the aquaria and in the sea were close because the seawater was pumped from a depth of 2 m of the fringing reef and was not subjected to filtration or settling. The water in the aquaria was intensively

aerated; water change was approximately 30% h^{-1}. The light intensity in open aquaria was 70–90% PAR_0 (high light). Some aquaria were shaded by gray or black plastic mesh. In the shaded aquaria, the light intensity was reduced to 30–50% PAR_0 (moderate light); 10–20% PAR_0 (low light) and 2–5% PAR_0 (extremely low light). Incident surface photosynthetic active radiation (PAR_0) was 1100–1400 μmol m^{-2} s^{-1}, 470–780 μmol m^{-2} s^{-1}, 160–320 μmol m^{-2} s^{-1}, and 30–80 μmol m^{-2} s^{-1} at midday during winter time and 800–1400 μmol m^{-2} s^{-1}, 600–1000 μmol m^{-2} s^{-1}, 200–400 μmol m^{-2} s^{-1}, and 40–100 μmol m^{-2} s^{-1} during summer time, in the respective four light regimes used.

2.2. Study specimens

Massive colonies of *Porites lutea* (Edwards and Haime 1860) and branched colonies of *Porites cylindrica* (Dana, 1846) were collected from open sites at the depth of 1–1.5 m from the fringing reef during low tide. The colonies were damaged by the bleaching events in 1998. Two types of fragments were cut from the colonies: (1) small pieces with weight ranging from 2 to 60 g each and (2) large fragments with weight ranging from 200 to 800 g each for *P. lutea* and from 80 to 200 g each for *P. cylindrica*.

2.3. Experimental design

At the beginning of the experiments, coral fragments were inflicted with injuries. Only one injury was made on each fragment.

Coral tissue was removed with Water-Pik (WP injury) from the top to the bottom of the large fragments and amounted to approximately ¼ part of whole surface area occupied with polyps. Such treatment led to removal of the live tissue from the skeleton. The fragments were also inflicted with injuries as follows: a chisel injury over the surface in the middle part of the fragment to remove superficial layer of tissue and skeleton 1–2 mm depth from the coenosteum. Breakage was simulated with pincers (for *P. cylindrica*) or with a small hand-saw (for *P. lutea*) after which the wounds were polished with a file (file injury). Osmotic shock (OS) injuries were made with a gentle jet of fresh water and live tissue was partly removed from the middle part of the lesions. The remaining tissue after subject to the osmotic shock died off and then detached from the skeleton. Healthy areas of some coral fragments were partly covered with a 1-mm thick layer of cement (cement injuries). Except for WP injuries, the lesions for all types of inflicted injuries were located in the middle part of the fragments. Injury size depended on the dimensions of the fragments and on average amounted no more than 10% of the living tissue area. After the procedure with injuries, all fragments were photographed and placed onto plastic grids in their natural position into the experimental aquaria.

2.4. Analytical procedures

2.4.1. Growth rate measurements

Coral fragments were blotted with a paper towel to remove excess seawater, weighed and placed back into the respective aquaria. The fragments were weighed again after 30 days of maintenance. The growth rate of samples was calculated using the formula:

$$\mu = (m_1 - m_0/m_0 \times \Delta T) \times 100;$$

where, μ is the average growth rate measured in mg g^{-1} day^{-1}, m_0 is initial weight, m_1 is the weight at the end of the experiment, and ΔT is the time between the two measurements of weight (Brinkhuis, 1985).

2.5. Recovery rate of lesions

To determine the amount of regeneration of the injuries over time, the injured areas of the experimental fragments were monitored photographically with digital camera "Olympus" C5050Z. The amount of regeneration within the experimental period was quantified from the color photographs by printing the image at 3× or 5× magnification and tracing the damaged and healed areas. For five coral fragments, distance from the center to the edge of lesions was measured immediately after the damage and then after one, two or some months of the lesions regeneration. Every time, 10 measurements were made for each fragment and the average regenerative rate (or the rate of living tissue spreading on newly formed substratum, mm per day) were

calculated on the basis of these measurements. Means and standard deviations were calculated on the basis of average data obtained for five fragments ($n=5$).

2.6. Amount of algal and animal organisms settled onto damage

To determine the area occupied by the algal and animal organisms, "weight" method was used. Color image of lesion was printed on high quality Epson KA450PM paper at 3× or 5× magnification. The image of the injured area was cut and weighed, and then the images of the algae and animals were similarly cut and weighed. The relative area occupied by the settlers was then calculated by the following formula:

$$\%S = W_2/W_1 \times 100;$$

where, S is the relative area of a projective cover of injury by settlers, W_1 is weight of paper with image of lesion, and W_2 is weight of paper with image of settlers. Means and standard deviations were calculated on the basis of five photographs ($n=5$).

Algal composition and abundance were determined under a stereoscopic microscope on the basis of 10 fortuitous optical fields.

2.7. Light intensity measurement

Light intensity was measured at the water surface in the open aquaria and near the upper parts of the coral fragments in both the open and shaded aquaria with a Li-Cor radiation sensor (Model LI-192 SB). The measurements were performed three times a day: at 9–10 h, 13–14 h and 17–18 h. Relative light intensity near the corals was calculated from these measurements and represented in percent of incident surface photosynthetic active radiation (PAR_0).

2.8. Statistical analyses

The raw data for regeneration and growth rates were arcsine transformed prior to use of analysis of variance (ANOVA) procedures. Tukey Honest Significance Test was employed for multiple comparisons of means whenever ANOVA results were significant. Differences were considered significant at $P<0.05$.

3. Results

3.1. Regeneration of different injury types and their colonization by settlers on P. lutea and P. cylindrica fragments during winter, spring and summer seasons

The greatest regeneration rate of tissue for all the injury types for both coral species occurred during winter (period I, Fig. 1). It dropped approximately 2-fold during spring (period II, Fig. 1) and then 2–3-fold during summer (period III, Fig. 1). During periods I and II, the regeneration rates of file injuries were approximately 2-fold and 4-fold higher than those of OS and cement injuries, respectively. During period III, the difference in the regeneration rates for various types of injuries was insignificant. Sometimes in summer, the recovered live tissue of the OS injuries was found bleached (Fig. 2A).

During winter and spring periods, the growth rate of the *P. lutea* fragments varied insignificantly and was approximately two times higher than during summer (Fig. 1C). There were no significant differences in the growth rate of *P. cylindrica* between fragments cultured at different seasons (Fig. 1D).

Means of the algal and animal settlement expressed as a percentage of the injury size for both coral species are presented in Table 1. During the first 60 days of the experiment, all the injury types were colonized mainly by algae (approximately 5–20% of the whole injured area). After 150 days of the maintenance, the overgrowth of chisel, file and WP injuries by algae amounted to 80–100% of the total injured area. The cement injuries were overgrown with the algae by almost 30%. The chisel and file injuries were occupied with animals about 2–3% of the injured area. On the 240th day of the experiment, chisel lesions healed whereas the OS, file and WP injuries were still overgrown by the algae that colonized about 90–100% of the total injured area. The cement injuries were occupied with the algae by about 60%.

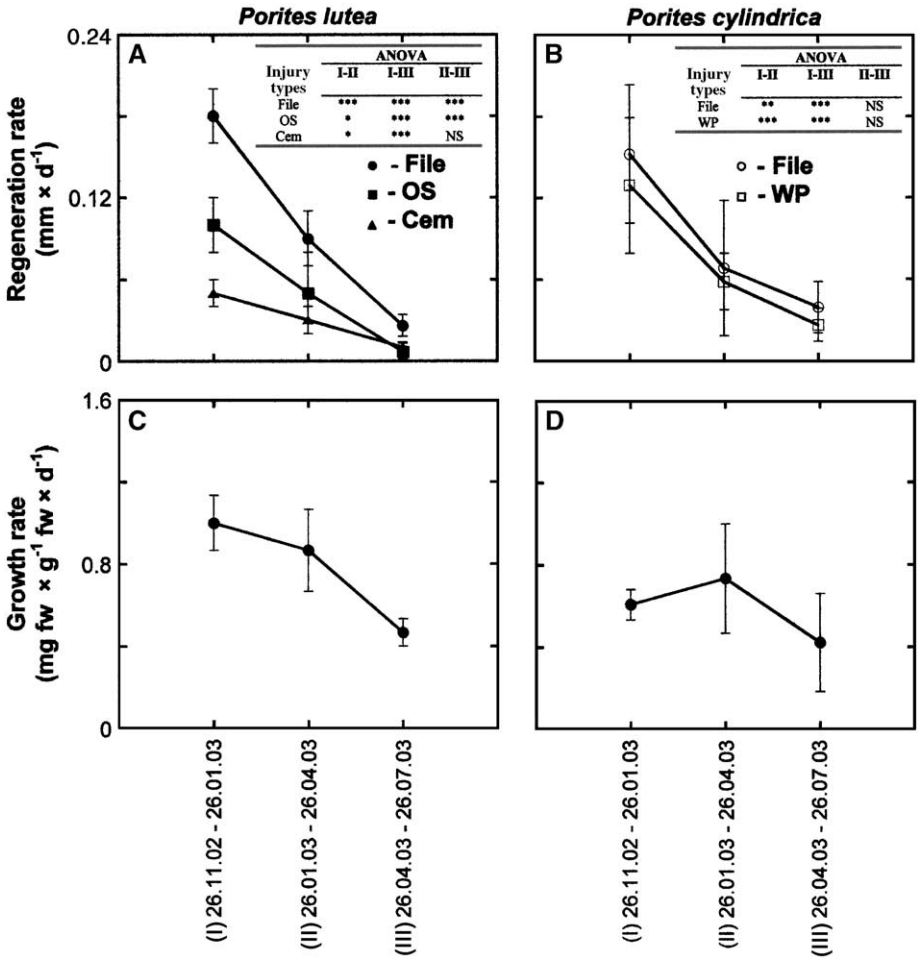

Fig. 1. Regeneration and growth rates of inflicted injuries on fragments of the corals *P. lutea* (A, C) and *P. cylindrica* (B, D) under light intensity 30–70% PAR_0 during winter, spring and summer periods. File—file injury; OS—osmotic shock injury; Cem—cement injury; WP—Water Pik injury. Error bars represent mean ± S.D. The inserts show the summary of statistical results for comparison of regeneration rates for different injury types among the three recovery periods (I—(26.11.02—26.01.03), II—(26.01.03–26.04.03) and III—(26.04.03–26.07.03)). Significant differences were tested by ANOVA analysis followed by Tukey Honest Significant Difference Test. $*P<0.05$, $**P<0.01$, $***P<0.001$. NS = not significant.

The list and abundance of the algae settled onto different types of injuries for both coral species are given in Table 2. In total, more than 100 algal species were found growing on the border areas of the file injuries of both coral species. During spring, the damaged areas of both coral species were mostly dominated by the green algae (Chlorophyta): *Enteromorpha clathrata*, *Cladophora* spp., *Ostreobium quekettii* and *Monostroma nitidum* (only on *P. cylindrica*), and by the brown algae (Phaeophyta): *Hincksia mitchelliae*, *Chilionema ocellata* and *Sphacelaria* spp. The red algae (Rhodophyta) were represented by *Centroceras clavulatum*, *Chondria minutula* and *Polysiphonia* spp. Among microscopic blue-green algae (Cyanobacteria), *Oscillatoria* spp. and *Calothrix* spp. were commonly found. *Nitzschia* spp. (Bacillariophyta) were found in mass among other algae. For marine animals, *Acervulina* spp. (Foraminifera) and *Spirorbis* spp. (Polychaete) were common settlers on the algae and dead surface of the corals.

During the summer period, the number of green, brown and red algae decreased, while algal

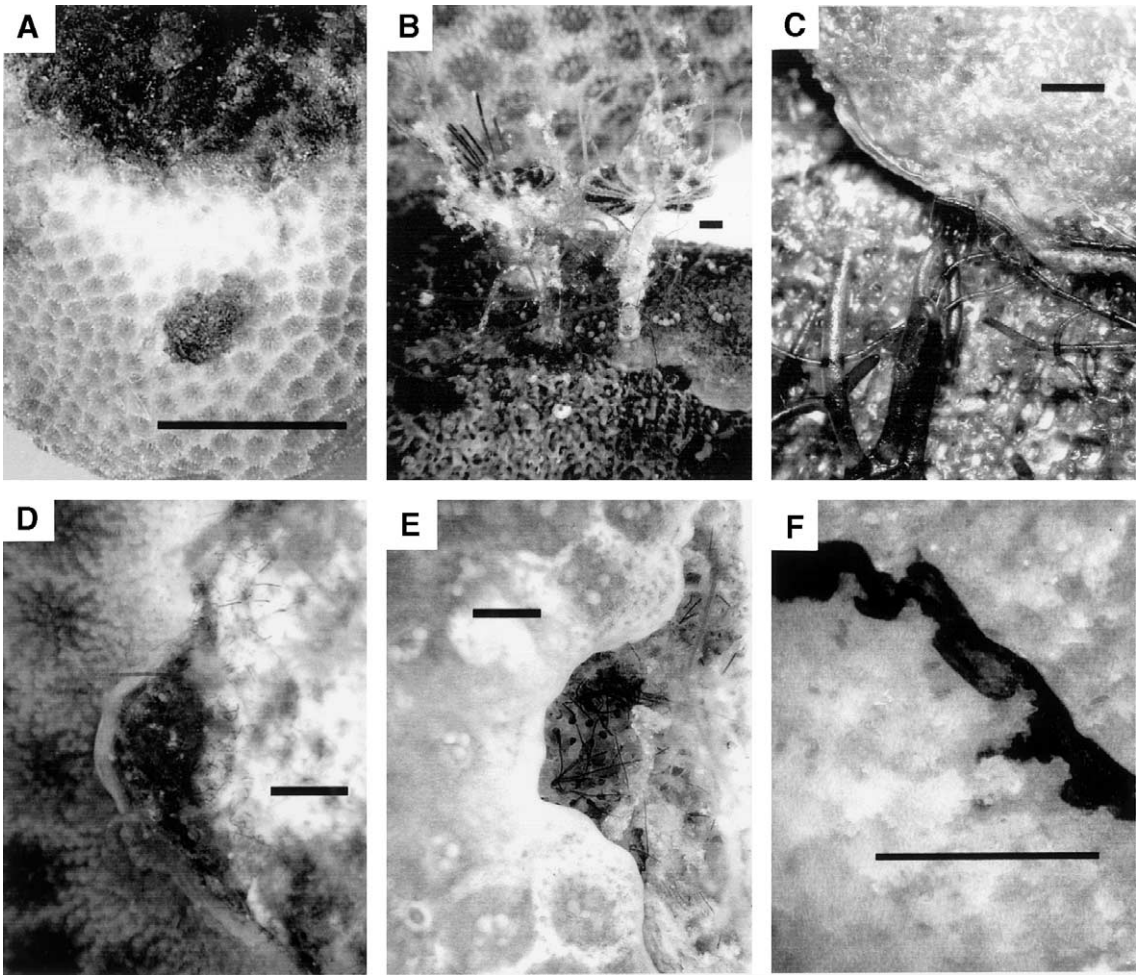

Fig. 2. (A) Bleaching and overgrowth of *P. lutea* polyps by the cyanobacterium *L. majuscula* colonized OS injury by the 120th day of the maintenance under 30–50% PAR_0 in the spring period. (B) Expansion of coral tissue around *Acetabularia* parvula stalk on file injury of *P. lutea*. (C) Overgrowth of turf algae (Coelothrix irregularis and *C.* clavulatum) by live tissue of the coral *P. cylindrica* on WP injury. (D) Overgrowth of entwined epilithic "felt" with sediments onto cement injury by live tissue of the coral *P. lutea*. (E) Impediment of the recovery of file injury on *P. lutea* by the colonizer *L. majuscula*. (F) Overgrowth of *Dictyota* sp. thallus on coral surface by the live tissue and entombing into calcium carbonate secret. For Fig. 1A, scale bar = 1 cm and for Fig. 1B–F, scale bars = 1 mm.

species of Cyanobacteria and Bacillariophyta increased. They were dominated by *Lyngbya majuscula*, *Oscillatoria* spp., *Phormidium* spp. and *Tolypothrix* spp. (Cyanobacteria), *Thalassia nitzschioides* (only on *P. lutea*), *Nitzschia longissima* (Bacillariophyta), *Derbesia marina* (only on *P. lutea*), *Ostereobium quekettii* (Chlorophyta), *H. itchelliae* (Phaeophyta), *C. clavulatum* and *Polysiphonia* sp. (Rhodophyta). For marine animals, *Acervulina* spp. (Protozoa) was the dominant species for both corals investigated. All algal species, except monodominant community of *L. majuscula* (Fig. 2A,E) that was found on the file injuries, were overgrown by the live tissue of both coral species (Fig. 2B,C,D,F).

3.2. Effects of light intensity on the recovery of injuries, coral growth rates and colonization by settlers

Under all light regimes, the regeneration rates of file and WP injuries were highest during the first

Table 1
Means and standard deviations of relative algal and animal cover (%) on injured areas of coral fragments maintained under moderate light intensity (30–70% PAR_0) during 60, 150 and 240 days

Coral species	Injury type	Settlers	From 26.11.02 to 26.01.03, 60 days	From 26.11.02 to 26.04.03, 150 days	From 26.11.02 to 26.07.03, 240 days
Porites lutea	chisel	algal	16.3 ± 2.1	80.6 ± 9.0	
		animal	<1	1.7 ± 1.5	
	OS	algal	8.3 ± 1.5	100	100
		animal	0	0	0
	cement	algal	7.0 ± 2.6	32.6 ± 11.3	52.0 ± 11.1
		animal	0	0	<1
Porites cylindrica	file	algal	19.7 ± 3.2	70.7 ± 11.6	87.0 ± 8.2
		animal	<1	2.0 ± 1.0	1.3 ± 0.6
	WP	algal	24.0 ± 9.8	100	100
		animal	0	0	0

month of the experiment (period I, Fig. 3A–D). There were no significant changes in the regeneration rates of the injuries in corals maintained under bright and low light (except file injuries on *P. cylindrica*, where the regeneration rate dropped twice under low light conditions). Under extremely low light, the regeneration rate dropped approximately three times in *P. lutea*, while it was almost not changed in *P. cylindrica*. During the following 2 months of exposure (period II), the regeneration rate of file injuries in both coral species sharply dropped (approximately four and two times for *P. lutea* and *P. cylindrica*, respectively) irrespective of the light treatment. The regeneration rate of WP injuries also dropped in both coral species during the period II.

In the coral fragments of *P. cylindrica* and *P. lutea* maintained under bright (70–90% PAR_0) and low (10–20% PAR_0) light intensities, the growth rates were approximately equal during the maintenance and amounted about 1 mg fw g^{-1} fw day^{-1}. Under extremely low light (2–5% PAR_0), it amounted to approximately 0.3 mg fw g^{-1} fw day^{-1} for both coral species (Fig. 3E,F). Colonization of artificial lesions by the algal and animal settlers under various light intensities is shown in Table 3. During the first month of the experiment, the algae occupied from 4% to 16% of the injured areas in the bright light treatment, from 6% to 14% under low light and only from 1% to 6% in the extremely low light treatments. The area occupied with animals during the first 30 days of the experiment was usually less than 1%. During the second experimental period (90 days), the subsequent algal colonization of the injuries averaged about 70–100% and 30–50% under bright and extremely low light, respectively. The area occupied by the animal organisms within 90 days of the experiment under bright light remained less than 1%. Under low and extremely low light, it varied from <1% to 6% and from 2% to 4%, respectively. There were no significant differences in the colonization by settlers between the file and WP injuries for both coral species.

The list of species and abundance of organisms settled onto the file injuries for both coral species maintained under different light intensities are shown in Table 2. Analyses of species composition of the algae settled onto file injuries carried out at the beginning of May (i.e. 3 months after the injuries were inflicted), showed significant differences in the composition and abundance of the algae living under various light intensities. Most of them (in total 102 species of algae) were found on the corals maintained under low light. Under bright light, 65 species and under extremely low light 43 species were found. Under bright light, the algal phyla such as Bacillariophyta, Cyanobacteria, Chlorophyta and Phaeophyta were dominant. Under low light, all the algal phyla were distributed approximately equally. Under extremely low light, members of the Rhodophyta and Phaeophyta were dominant. Under bright light, the algae such as *N.*

Table 2
Species list and abundance of algae and animals settled onto the corals *Porites lutea* and *Porites cylindrica* after damages (file injuries)

Species list	I*		II*		III*		
			30–70% xPAR$_0$		70–90% PAR$_0$	10–20% PAR$_0$	2–5% PAR$_0$
	P. lutea	*P. cylindrica*	26.04.03	26.07.03	06.05.03		
1	2	3	4	5	6	7	8
Bacillariophyta							
Fragilaria sp.	+		+		+	+	+
Asterionella japonica Cleve	++	××	++/××	+/××		+/××	++
Thalassionema nitzschioides Grunow?	+++		++	+++		+++	
Diatoma sp.	+++	××	+/××	+	××	+/××	+++
Navicula sp.	+	×	+	+/×	×	+/×	×
Nitzschia closterium (Ehrenberg) W. Smith	++	×	++/×	+	++/×	++	
Nitzschia longissima (Brebisson) Ralfs	+++	×××	+++/×××	++/×××	+++/×××	+++/×××	
Nitzschia seriata Cleve	++	×××	++/×××	×	++/×	++/×××	×
Licmophora abbreviata Cleve	+++	×	++/×	+/×	++	+++/×	
Cyanobacteria							
Dermocarpella clavata (Setchell & Gardner) Pham Hoang Ho	+	××	+	+		+/××	
Dermocarpa acervata (Setchell & Gardner) Pham Hoang Ho	+	××	+/××	+	×	+/××	
Aphanocapsa littoralis Hansgirg	+	××	+	+/××	×	+/××	×
Lyngbya epiphytica Hieron	+	×	+/×	+	×	+	
Lyngbya majuscula (Dillwyn) Harvey	+++	×	+/×	+++/×	+++/×	+++	
Lyngbya martensiana Meneghini ex Gomont	+	×	+	+/×	×	+	
Lyngbya polychroa (Meneghini) Rabenhorst [*Lyngbya sordida* (Zanardini) Gomont]		×	×		×	×	
Lyngbya spp.	+	×	+/×	+	+	+/×	×
Oscillatoria margaritifera (Kutzing) Gomont	+	×	+/×	+/×		+/×	
Oscillatoria miniata (Zanardini) Gomont	+++	×	+++/×	+/×	×	+++/×	
Oscillatoria tenuis Agardh	++	×	+/×	+		++/×	
Oscillatoria sp.	+++	×	+++/××	+++/×	+/×	+++/××	
Phormidium corium (C. Agardh) Gomont	+	××	+	+/×	×	+/×	×
Phormidium tenue (Meneghini) Gomont	+	×	+/×	+/	+/×	+/×	
Phormidium sp.	+++	×	+	+++/×		+++/×	×
Spirulina subsalsa Oersted ex Gomont	+	×××	+/×××	+/××	×××	+/×	×
Spirulina subtilissima Kützing	+	×		+/×		+/×	
Spirulina tenerrima Kützing	+	×	+/×	×	×	+/×	
Spirulina sp.	+++		+++			+++	
Calothrix confervicola (Roth) Agardh	+	×	+/×	×		+/×	
Calothrix contarenii (Zanardini) Bornet & Flahault	+	×	+/×	×	×	+/×	×
Calothrix scopulorum (Weber et Mohr) C. Agardh	+	×	×	+		+/×	
Calothrix spp.	+++	××	+++/××	+/×	+++/×	+++/××	×
Dichotchrix sp.	++	×	+/×	×	+/×	++/×	+/×
Tolypothrix sp.	+++	×	+	+++/×		+++/×	×
Chlorophyta							
Acrochaete viridis (Reinke) R. Nielsen	++		++	++	++	++	
Monostroma nitidum Wittrock	+++		+++		+++	+	
Enteromorpha clathrata (Roth) Greville	+++	×××	+++/×××	+	+++/×××	++/××	×
Enteromorpha ralfsii Harvey?	+	×	×		+	×	+
Enteromorpha sp.	+	×	+/×	+	+/×	+/×	+/×
Ulva sp.	+	×	+/×			+/×	

Table 2 (*continued*)

Species list	I*		II*		III*		
			30–70% xPAR$_0$		70–90% PAR$_0$	10–20% PAR$_0$	2–5% PAR$_0$
	P. lutea	*P. cylindrica*	26.04.03	26.07.03	06.05.03		
1	2	3	4	5	6	7	8
Chlorophyta							
Pringsheimiella scutata (Reinke) Höhnel ex Marchewianka	+	×	+/×		+/×	+/×	
Rhizoclonium implexum (Dillwyn) Kützing	+		+			+	
Cladophora laetevirens (Dillwyn) Kützing		×	×			×	
Cladophora vagabunda (Linnaeus) Van den Hoek	+	×	+/×			+/×	
Cladophora sp.	+	××	+/××	+	+/××	+/×	×
Cladophoropsis sp.	+	××	+		+	+/××	
Dictyosphaeria cavernosa (Forsskål) Børgesen		×	×		×	×	
Boodlea composita (Harvey) F. Brand	+		+	+	+	+	
Derbesia fastigiata W.R. Taylor		×	×		×	×	
Derbesia marina (Lyngbye) Solier	+++		+	+++	+++	++	
Codium sp.		×	×		×		
Rhipidosiphon javensis Montagne [*Udotea javensis* (Montagne) A. Gepp & E. Gepp]	+++	×			++	+++/×	
Ostreobium quekettii Bornet et Flahault	+++	×××	+++/×××	+++	+++/××	++/×××	+++/×
Penicillus sibogae A. Gepp & E. Gepp	+			+		+	
Neomeris annulata Dickie		×	×		×	×	
Acetabularia dentata Solms-Laubach		×	×		×	×	
Acetabularia exigua Solms-Laubach		×	×		×	×	
Acetabularia parvula Solms-Laubach		×	×		×	×	
Phaeophyta							
Ectocarpus elachistaeformis Heydrich	+	××	+/××		×	+/×	×
Ectocarpus sp.	++	×	++/×	+	+/×	+/×	×
Feldmannia irregularis (Kützing) G. Hamel	++	×	++/×	+	++/×	+/×	
Hincksia mitchelliae (Harvey) Silva in Silva et al.	+++	×××	+++/×××	+/××	+++/×××	++/×××	×
Hincksia sp.	+	×	+			+/×	
Pilayella littoralis (Lamouroux) Kjellman	+		+			+	
Chilionema ocellata (Kützing) Kuckkuck	++	××	++/×		+/×	++/×	××
Sphacelaria novae-hollandiae Sonder	+++	×	+++/×	+	++/×	+++/×	++
Sphacelaria rigidula Kützing	+++	×	+++/×	+/×	++/×	+++/×	++/×
Sphacelaria tribuloides Meneghini	++	×	++/×	++/×	×	++/×	×
Sphacelaria sp.	+		+			+	
Lobophora variegata (Lamouroux) Womersley	+		+			+	
Dictyota linearis (C. Agardh) Greville		×	×			×	
Dictyota sp.		×					×
Chnoospora implexa Hering ex J. Agardh		×	×		×	×	
Rhodophyta							
Chroodactylon ornatum (C. Agardh) Basson		×	×		×		
Stylonema alsidii (Zanardini) K.M. Drew	+	×	+/×	+	×	+/×	
Erythrotrichia carnea (Dillwyn) J. Agardh	+	×	+/×			+/×	
Audouinella virgarula (Harvey) Dixon	+	×	+/×	+		+/×	×
Audouinella sp.	+	×	+/×	+	+/×	+/×	×
Hydrolithon farinosum (J.V. Lamouroux) Penrose & Y.M. Chamberlain	+	×	+/×	×		+/×	×
Peyssonnelia inamoena Pilger		×				×	×

(continued on next page)

Table 2 (*continued*)

Species list	I*		II*		III*		
			30–70% xPAR$_0$		70–90% PAR$_0$	10–20% PAR$_0$	2–5% PAR$_0$
	P. lutea	*P. cylindrica*	26.04.03	26.07.03	06.05.03		
1	2	3	4	5	6	7	8
Rhodophyta							
Pneophyllum fragile Kützing	+	×	+	+/×		×	+
Sporolithon sp.	+		+	+	×	×	+
Champia parvula (C. Agardh) Harvey		×	×				
Gelidiella pannosa (Feldmann) Feldmann & Hamel	+		+			+	
Jania capillacea Harvey	+		+			+	
Falkenbergia hillebrandii (Bornet) Falkenberg	+		+			+	
Anotrichium tenue (C. Agardh) Nägeli			+	+	+	+	
Antithamnion lherminieri (P. Crouan & H.Crouan) Bornet		×	×				
Centroceras clavulatum (C. Agardh) Montagne	++	××	++/×	++/××	+/×	++/×	+/×
Centroceras inerme Kützing		×	×			×	
Ceramium fastigiatum Harvey	+	×	+	+/×		+/×	
Ceramium flaccidum (Harvey ex Kützing) Ardissone	+	×	+/×	+	×	+/×	
Ceramium sp.	+	×	+/×		+	×	+/×
Corallophila apiculata (Yamada) R. Norris	+	×	+/×	+		+/×	
Crouania attenuata (C. Agardh) J. Agardh		×			×	×	×
Spyridia filamentosa (Wulfen) Harvey	+	×	×	+/×		+/×	×
Chondria dasyphylla (Woodward) C. Agardh	+		+		+		
Chondria minutula Weber-van Bosse	+	×	++/×			+/×	
Chondria sp.	++	×	+			+	×
Spermothamnion sp.	+	×	+		+	+	×
Herposiphonia secunda (C. Agardh) Ambronn f. *secunda* Wynne	+	×	+	×	×	+/×	×
Herposiphonia secunda (C. Agardh) Ambronn f. *tenella* Wynne		×	×		×	×	
Herposiphonia sp.	+			+		+	
Laurencia implicata J. Agardh		×	×			×	
Laurencia obtusa (Hudson) J.V. Lamouroux	+	×	+	+/×		+/×	×
Laurencia sp.		×	×			×	×
Polysiphonia sp.	+++	×××	+++/×	+/×××	+++/×	+/×	×××
Tolypiocladia glomerulata (C. Agardh) F. Schmitz		×	×			×	
Protozoa							
Peridinium sp.	+			+		+	
Foraminifera sp.		××	××			+	
Acervulina sp.	++	××	++/××	++/×	++/×	+/×	+
Peneroplis planatus	+			+		+	
Urochordata							
Tunicate sp. (solitary tunicates)	++	××	+/×		××	++/×	
Polychaete							
Spirorbis sp.	+++	××	+/××	×		+++/××	+++/××

I*—abundance of settlers on file injuries of *P. lutea* and *P. cylindrica* for all experiments. II*—abundance of settlers on file injuries of *P. lutea* and *P. cylindrica* maintained in aquaria under moderate light intensity (30–50% PAR$_0$) from 26.11.02 to 26.07.03. III*—abundance of settlers on file injuries of *P. lutea* and *P. cylindrica* maintained in aquaria under different light intensities from 26.02.03 to 06.05.03.
+—for *P. lutea* and ×—for *P. cylindrica*; +++/×××—for more than 10 individuals per cm^2; ++/××—for less than 10 individuals per cm^2; +/×—for less than 1 individual occurred per cm^2.

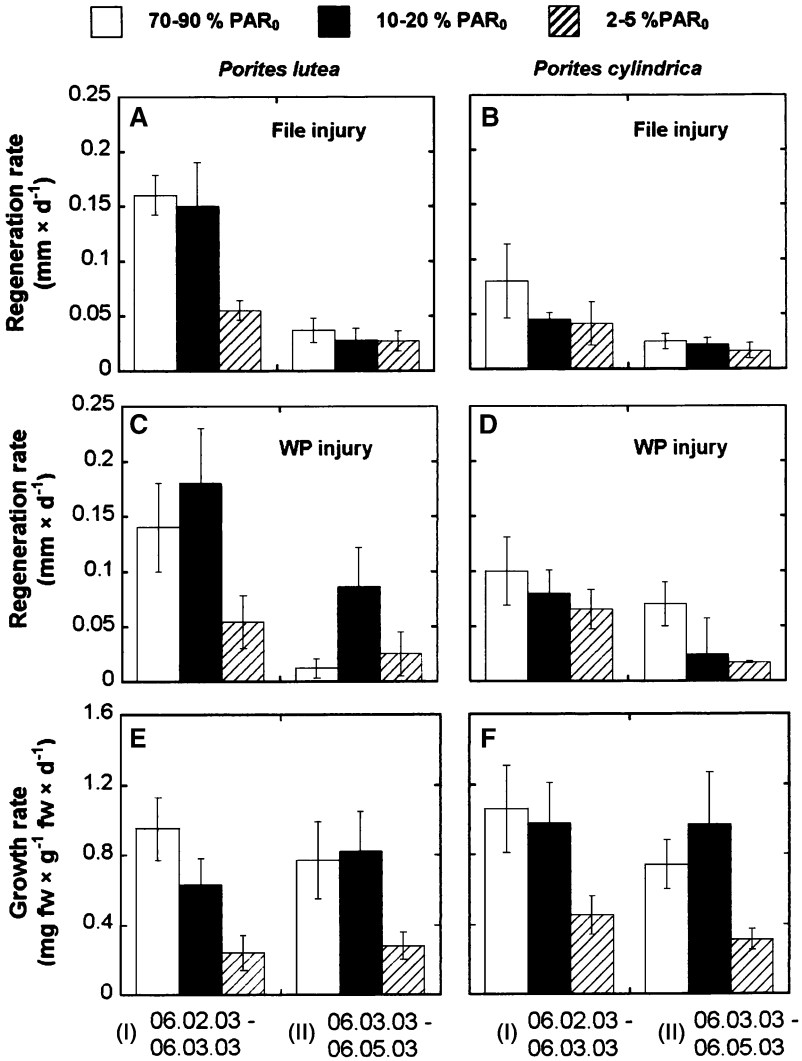

Fig. 3. Regeneration and growth rates of damaged tissue in fragments of the corals *P. lutea* (A, C, F) and *P. cylindrica* (B, D, F) at different irradiance during winter and spring periods. Error bars represent mean ± S.D.

longissima (Bacillariophyta), *L. majuscula*, *Spirulina subsalsa* (only on *P. cylindrica*) and *Calothrix* spp. (Cyanobacteria), *M. nitidum* and *D. marina* (only on *P. lutea*), *E. clathrata*, *O. quekettii* (Chlorophyta), *H. mitchelliae* (Phaeophyta), *Polysiphonia* spp. (Rhodophyta) were common. Under low light, these species were also found in mass quantity (except for *Polysiphonia* sp.). Moreover, high density of the algae such as *Thalassionema nitzschioides* (only on *P. lutea*), *Nitzschia seriata*, *Licmophora abbreviata* (Bacillariophyta), *Oscillatoria* spp., *Phormidium* spp., *Spirulina* spp. (only on *P. lutea*) and *Tolypothrix* spp. (Cyanobacteria), *Rhipidosiphon javensis* (Chlorophyta), *Sphacelaria* uhaeophyta), *C. clavulatum* (Rhodophyta) were detected. Under extremely low light, high population density was recorded for *Diatoma* spp. (Bacillariophyta) (only for *P. lutea*), *O. quekettii* (Chlorophyta), *Sphacelaria rigidula* (Phaeophyta) and *Polysiphonia* spp. (Rhodophyta). For animals,

Table 3
Means and standard deviations of relative algal and animal cover (%) on injured areas after 30 and 90 days maintenance in aquaria under different light regimes

Coral species	Injury type	Light intensity %, PAR_0	From 06.02.03 to 06.03.03, 30 days		From 06.02.03 to 06.05.03, 90 days	
			Algae	Animals	Algae	Animals
Porites lutea	File	70–90	7.7 ± 2.5	<1	71.6 ± 7.6	<1
		10–20	8.3 ± 1.5	<1	88.3 ± 7.6	6.3 ± 1.5
		2–5	6.0 ± 2.0	<1	51.3 ± 12.0	2.7 ± 1.5
	WP	70–90	3.7 ± 2.0	0	100	0
		10–20	14.0 ± 2.6	0	89.6 ± 5.5	<1
		2–5	0.83 ± 0.28	<1	32.3 ± 7.8	1.3 ± 0.6
Porites cylindrica	File	70–90	16.6 ± 5.0	<1	82.6 ± 10.4	<1
		10–20	5.7 ± 3.2	1.2 ± 0.3	73.0 ± 8.0	3.7 ± 2.08
		2–5	2.3 ± 0.6	<1	27.6 ± 10.6	4.3 ± 1.15
	WP	70–30	12.0 ± 10.6	<1	100	0
		10–20	11.3 ± 1.5	1.3 ± 0.6	74.0 ± 8.8	3.0 ± 1.7
		2–5	1.7 ± 0.7	3.7 ± 2.1	36.0 ± 9.1	2.2 ± 0.3

Spirorbis spp. (Polychaete) was dominant under low and extremely low light.

4. Discussion

4.1. Dynamics of repair of the damaged coral tissue and competitive interactions between the corals and the algae during various seasons

As was shown in the previous investigations (Bak, 1983; Meesters et al., 1992, 1994; Hall, 2001) and in our experiments, regenerative ability of injured corals in due course decreased under all the tested conditions. One of the main reasons of such an event occurring during the first months of the recovery might be the difference in the rates of the recovery of damaged polyps and the formation of new polyps. As considered by Hall (2001), this fact could be explained by the energetic expenses for these processes, i.e. it is energetically more costly to produce new polyps than to repair partially damaged ones. Our experiments showed that re-growth of the partially damaged polyps occurred during the first month of the experiment and then, new polyps started to form. The regenerative rate of the partially damaged polyps was faster than the formation of new polyps. This fact was confirmed by the differences in the rates of regeneration between different injury types during the experiments. During the first 2 months, the repair of the file injuries in *P. lutea* was two and three times higher than that of WP and cement injuries, respectively. Our estimations showed that the area of the partially damaged polyps amounted to 10–20% in the file injuries, 5–10% in the WP injuries and about 1% of the whole area of the damages in the cement injuries. Bak and Steward-van Es (1980) have reported similar results for the recovery of scraping and tissue injuries for *Porites asteroids*. The authors have shown that the rapid recovery observed was assisted by the regeneration of the partially injured polyps within the injured site.

Interspecific differences in the rates of regeneration for all the injury types were mostly found within the first month of the experiment. The recovery of file and WP injury types in *P. lutea* was approximately two times faster than that in *P. cylindrica* both under bright and low light conditions. This fact undoubtedly appertains to the differences in the structure of these corals. The type of injury as well as the structure (morphology) of the polyps effect the number of partially damaged polyps. Dimension of the file injury restricted mainly by the polyp sizes and the height of corallites (i.e. on depth of the spread of live tissue). The distance from the breakage edge (distal margin of lesion) to the base of damaged polyps (corallites bottom) was approximately two times longer in *P. lutea* than in *P. cylindrica*. Consequently, partly damaged tissue with the same injury type inflicted (file or WP) in *P. lutea* had a greater area of the lesion than that in *P. cylindrica*.

Regenerative ability of injured colonies with different morphologies has been studied in eleven coral species (Hall, 1997). The regenerative abilities could be ranked according to morphological attributes for these species: arborescent > bushy > tubular > massive > submassive. According to the scheme, recovery of injuries for *P. cylindrica* (bushy) must be faster than that for *P. lutea* (massive). However, we did not obtain the confirmation for that in our experiments. A plausible explanation for this might be the differences in the physiological state of the corals and conditions under which the corals were maintained in the present and Hall's experiments.

At the first stage of the regeneration of damaged polyps, coral/algal competition probably does not determine the recovery rate, although it takes place. Damaged tissue of the corals *P. cylindrica* and *P. lutea* always overgrew algal spores and fragments settled onto the surface of lesions and then entombed them into the skeleton (Fig. 2F).

Long-term experiments showed that the regeneration rate for all injury types dropped 2–3 times within summer compared to spring. During these periods, lesion healing was occurred via formation of new polyps and occupation of dead skeleton overgrown with algae. Two explanations could be proposed for the reduction in the rates of wound healing during the summer period. Firstly, the reduced regeneration rate depended on 1.5–2-fold decrease in the growth rates of the corals probably caused by elevated temperature in aquaria of up to 30–31 °C at midday during summer. Secondly, the algal and animal settlers onto new substrate caused the slower regeneration.

The first supposition is confirmed by measurements of growth rates. Mass of the coral fragments significantly (two times) dropped in *P. lutea* during summer. The second supposition is confirmed by data on measurements of injured area colonized by the algae and the animals. For instance, algal coverage on cement injuries amounted 7% and 32.6% in winter and spring seasons, respectively, whereas it reached 52% in summer season. In most cases, algal communities act as a physical impediment for coral live tissue extension on available substrate (Fig. 2A,B,C). Moreover, some of them also act as a chemical impediment (Fig. 2E) probably because of the release of secondary products which could poison the coral tissue. The algal composition changed significantly from spring to summer: the number of algae dropped by 40 species. However, new species appeared during the same period and some of them such as *L. majuscula* and *L. bouillonii* (Cyanobacteria) poisoned the corals (Fig. 2A,E; Titlyanov et al., 2004).

A previous investigation by Jompa and McCook (2002) has shown that the brown alga *Lobophora variegata* caused tissue mortality in *P. cylindrica*, apparently by adherently overgrowing the basal part of the coral branches upward, and consequently smothering and killing the underlying coral tissue. McCook et al. (see review, 2001) have described many examples of overgrowing of corals by the algae such as *Dichotchrix* and *Sphacelaria*, *Cladophoropsis*, *Cladophora* and *Pneophyllum*. However, in our experiments, all the abovementioned algal species were found on the injured corals and were overgrown by the corals. Moreover, the algae did not display any negative effects (e.g., necrosis) on the recovery of coral tissue, although, they impeded the tissue growth via physical obstacles. Recent studies (Jompa and McCook, 2003a,b) have reported that the red algae *Corallophila huysmansii* and *Anotrichium tenue* were able to colonize and overgrow live tissue of the branching coral *P. cylindrica* and that the live tissue became bleached and swollen and subsequently died. In our experiments, *A. tenue*, which was rarely found and occurred only on damaged *P. lutea*, did not present any serious impediment for the coral. *C. huysmansii* was not recorded on damages. Tanner (1995) and Miller and Hay (1998) also have found relatively minor effects of primarily upright macroalgae on the coral mortality. Several studies (Meesters and Bak, 1993; Meesters et al., 1994, 1997a,b; Van Woesik, 1998; McCook, 2001) have shown that filamentous turfs are poor competitors with corals in the field as well as under experimental conditions. Two recent works (Jompa and McCook, 2003a,b) also have shown that mixed algal turfs were relative poor competitors with *P. cylindrica*.

Thus, our experiments showed that, when the corals *P. cylindrica* and *P. lutea* were recovered from injuries, these species overgrew more than 100 species of micro- and macro algae, and only two species of the Cyanobacteria (*Lyngbya bouillonii* and *L. majuscula*) acted as poison for the corals and significantly inhibited their growth, lesion healing and sometimes bleached the corals (Titlyanov et al., 2004).

We suppose that allelopathic effect of *L. majuscula* and *L. bouillonii* on live coral tissue was probably caused by biologically active metabolites such as curacin D, barbamide, dechlorbarbamide (found in *L. majuscula*) that are toxic for many marine organisms (Marquez et al., 1998; Chang et al., 2002; Wu et al., 1997; Gerwick et al., 2003). For example, curacin D is an antimitotic agent (Marquez et al., 1998), which probably inhibits nuclei division in polyp cells as well as in their zooxanthellae. Previously, De Nys et al. (1991) have shown that the red alga *Plocamium hamatum* could affect and kill live tissue of the octocoral *Sinularia cruciata* by releasing the secondary product—Chloromertensene.

4.2. Ability to recover from injuries under maintenance of fragments in aquaria with different light intensities

Recovery from injuries had similar rates for both coral species under bright (70–90% PAR_0) and low (10–20% PAR_0) light and dropped only under extremely low light (2–5% PAR_0), which is close to the light limiting levels for scleractinian coral distribution across depth (Titlyanov and Titlyanova, 2002). Moreover, wound healing and new polyps formation at the injured areas for *P. lutea* under light intensity less than 1% PAR_0 was detected (Titlyanov et al., unpublished data). It is assumed that regeneration of injuries has priority trend in metabolism of corals even under extreme conditions. Probably, scleractinian corals are able to use the energy of heterotrophic metabolism as a result of predation of zooplankton or digestion of its own zooxanthellae (Titlyanov et al., 1996, 2000) for tissue recovery under low light.

Biomass and structure of the community of algal settlers affected the rate of live tissue extension on coral lesions. Under bright and low light, algae covered the injured area approximately by 80–90% within 90 days. The number of algal species found on injured areas of the corals maintained under low light was higher than that under bright light. Under these both light intensities, *N. longissima* (Bacillariophyta), *L. majuscula* (Cyanobacteria), *E. clathrata* (Chlorophyta) and *H. mitchelliae* (Phaeophyta) were dominant. Algal density and size of mass algal species were similar under bright and low light exposures. Thus, under the conditions of bright and low light, the algae (and animals) settled onto dead surfaces of the damaged corals caused approximately equal physical and chemical impediments for the spreading of live tissue of *P. cylindrica* and *P. lutea*. The situation was sharply changed under extremely low light. The projective cover by algae amounted only to 20–40% of injured area and the number of algal species dropped two times in the extremely low light treatment compared to the low light treatment. The poisonous cyanobacterium *L. majuscula* was absent. The number of animals (especially *Spirorbis* spp.) was approximately the same with that observed under low light. Thus, algal colonizers on damaged corals acted as physical and chemical (allelopathic) impediments for recovery of polyps under a wide range of light intensity (from 2% to 90% PAR_0). Mass and morphology of the algae, their density and negative allelochemical properties affected the regenerative ability of the corals.

5. Conclusions

Two species of scleractinian corals, the massive *P. lutea* and the branching *P. cylindrical*, were able to completely regenerate tissue and skeleton from injuries with partial or complete tissue removal. The injured areas healed in two stages. At the beginning, partially damaged polyps healed and then new polyps were formed on the injured areas. During the first stage, the injury type, morphology and anatomy of the injured tissue, physiological state of the corals, light conditions and to a lesser degree coral/algal competition affected the duration of the polyp regeneration. During the second stage of the recovery, the rate of live tissue extension and new polyp formation on dead areas of lesions were mainly affected by physiological state of the damaged coral, light and temperature during their maintenance and by coral/algal competition. Competitive interactions between the newly formed polyps and the algal/animal settlers took place only under their direct contact. Both coral species were able to overgrow more than 100 species of the algae and some species of sessile animals on the dead areas of the injuries. All settlers acted as physical impediment for the spreading of live coral tissue and

thus reduced the rate of occupation of the substrate by the live tissue. Only in one case, allelopathic effect of the algae on corals was detected: mono-dominant community of *L. majuscula* (Cyanobacteria) being under direct contact with polyps of both coral species, inhibited coral growth and even bleached the polyps.

Thus, results of the experiments conducted under conditions close to natural environments suggest that scleractinian corals at the fringing reef of Sesoko Island successfully recover after the 1998 coral bleaching. Algae and animals settled onto damaged areas impeded the recovery of coral reef by reducing the rate of occupation of the substrate by the polyps. Some ephemeral algae (e.g., *L. majuscula*) temporally impeded the recovery due to allelopathic effect. The average rate of lesion healing (about 4 mm/year) supports the idea that corals damaged during the 1998 bleaching events can completely recover after 30–50 years under favorable conditions (i.e. in the absence of negative effects of natural catastrophes and pollution due to human activities).

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